Unit5 Market failure and government intervention in markets

5.1 meaning of market failure

- Market failure as a result of misallocation of resources
  - Market failure occurs when the market forces of supply and demand result in an inefficient and/or inequitable allocation of resources
- meaning of a misallocation of resources
  - resources are not put to their best or most effective use
  - Welfare is not maximised
  - more or less of the product should be produced and/or it needs to be allocated to consumers
  - left to the free market
    - price may be too high, too low or fluctuate too much
- Situations that can lead to market failure
  - public goods
  - externalities
  - merit and demerit goods
  - monopoly
  - other market imperfections
  - inequalities in income and wealth

5.2 Private goods, public goods and quasi-public goods

- Non-rivalry, non-excludability and their significance
  - Non-rivalry in consumption
    - consumption of a product by one person does not reduce amount available to others
      - eg. street light
  - Non-excludability
    - non-payers cannot be prevented from using the product
  - Significance of these characteristics and other examples
    - unwilling or unable to pay
    - cannot determined how much has been consumed
    - have a little chance to make profit for producers
- Public goods and private goods
  - Public goods
    - non-excludable and non-rival in consumption
  - Private goods
    - rival in consumption and excludable
- How a public good may become a quasipublic good
  - when a public good has some characteristics of a private good
  - The significance of technological change
    - encryption
- The free-rider problem
  - problem of non-excludability leads to the free-rider problem
    - where people can consume as much as they like of a good, without having to pay for it
- The tragedy of the commons
  - free-rider problem can lead to tragedy of the commons
    - where individuals act in their own self-interest and this results in the depletion (using up) or degradation (damage) of a shared resource.

5.3 Positive and negative externalities in consumption and production.

- Meaning and types of externalities
  - Externalities
    - effects on other people or firms (third parties) due to the consumption or production of goods and services.
    - may be the result of
      - consumption
      - production
    - can be
      - positive
      - negative
  - consumption externalities
    - effects on third parties due to the consumption of goods and services
  - production externalities
    - effects on third parties due to the production of goods and services
  - positive externalities/external benefits
    - benefits to third parties, for which they do not have to pay
  - negative externalities/external costs
    - costs to third parties, for which they do not receive compensation
  - Divergence between private and social costs and benefits
    - private costs
      - the costs to those involved in a transaction, the costs to the consumers and firms involved
    - social costs
      - the costs and any harmful effects to those involved in the transaction and to any third parties affected
      - SC = PC + EC
    - private benefits
      - the benefits to those involved in a transaction, the costs to the consumers and firms involved
    - social benefits
      - the benefits and any beneficial effects to those involved in a transaction and to any third parties affected
      - SB=PB+EB
    - If there are no external benefits, the social benefits will be the same as the private benefits
    - If there are no external costs, the social costs will be the same as the private costs
- The effects of positive and negative externalities
  - negative externalities in production
    - Sm is the supply curve when firms only consider their private costs
      - If there are negative externalities in production
        - social costs > the private costs
          - Ss shows the supply curve taking account of the full social costs
            - Left to the market, the equilibrium price is pm and the quantity sold is qm
    - If the firms have to pay for the external costs as well as the private costs
      - produce less and charge a higher price
        - equilibrium price will be ps where qs is sold
          - social optimum: the quantity that maximises social welfare, and reflects social costs and social benefits of an activity
    - left to the market
      - misallocation of resources
        - too much being produced at too low a price
          - negative externalities in production
            - overproduction
  - negative externalities in consumption
    - Dm is the demand curve when individuals only consider their private benefits
      - Ds is the demand curve taking account of the social benefits
        - because no consideration of the negative effects on third parties
          - market demand > demand that is best for society
            - If there are negative externalities in consumption
              - private benefits > social benefits
                - too much demand
                - price and output that are both too high
                - if left to the market
                - price will be pm and the quantity consumed will be qm
    - If individuals considered the benefits for society as a whole
      - quantity consumed and the price will be lower
        - qs and (psisocial optimum)
    - If left to the market
      - too much being consumed
        - negative externalities in consumption
          - overconsumption and hence overproduction
  - positive externalities in consumption
    - Dm is the demand curve when individuals only consider their private benefits
      - Ds is the demand curve taking account of the social benefits
        - because no consideration of the positive effects on third parties
          - market demand < demand that is best for society
            - If there are positive externalities in consumption
              - social benefits > private benefits
                - not enough demand
                - price and output are both too low
                - left to the market
                - pm and qm
    - If individuals considered the benefits to the society as a whole
      - quantity consumed and the price will be higher
        - qs and ps
          - social optimum
    - left to the market
      - too little being consumed
        - underconsumption and hence underproduction
    - also possible
      - product that results in positive externalities in consumption also generates negative externalities in production
        - whether there is underproduction will depend on which of the externalities is greater
  - positive externalities in production
    - Sm is the supply curve when firms only consider their private costs
      - positive externalities in production
        - private costs > social costs
          - Ss shows the supply curve taking account of the social costs
            - left market
              - firm that developed the technology does not consider the lower costs of others
                - pm and qm
    - If firms took account of the true costs to society
      - produce more and charge a lower price
        - ps and qs
          - social optimum
    - misallocation of resources
      - too little being produced at too high a price
        - positive externalities in production
          - underproduction
- the absence of property rights may result in externalities and market failure
  - Property rights
    - the authority to determine how resources is used, the ownership of land, sea air and any resources on or within it
    - Property Rights and Market Failures
      - Lack of clearly defined property rights can lead to market failures
        - Overuse and Depletion
          - tragedy of common
        - Underinvestment
          - people are uncertain about their property rights
            - less likely to invest in improving/maintaining their property
        - Inefficiency
          - Ambiguous property rights
            - resource allocation that does not maximize the welfare of society
    - Intellectual property rights: the ownership of a creation of someone's mind.
      - lack of IPR
        - another person or firm to copy or sell someone else's idea or product
          - less incentive to do so if creators were unable to earn enough profit
            - fewer new products and ideas being developed
              - inefficient

5.4 Merit and demerit goods

- Value judgements and the classification of merit and demerit goods
  - Merit and demerit goods are examples of partial market failure
  - merit good
    - a product that society or government judges desirable and too little is provided by the market for the benefit of society as a whole
  - demerit good
    - a product that society or government judges is undesirable and too much is provided by the market for the benefit of society as a whole
  - what are merit or demerit goods will vary between countries, according to moral or political judgements
- Merit and demerit goods, and positive and negative externalities in consumption
  - Merit goods and positive externalities in consumption
    - Not all products that result in positive externalities in consumption are merit goods
  - Demerit goods and negative externalities in consumption
    - Not all products that result in negative externalities in consumption are demerit goods
- How imperfect information can affect provision of merit goods and demerit goods
  - not know the full or long-term effects of actions they take now
  - focusing on short-term satisfaction or utility
  - merit goods tend to be underconsumed and underprovided
  - demerit goods tend to be overconsumed and overprovided

5.5 Market imperfections

- imperfect and asymmetric information can lead to market failure
  - Imperfect information exists when buyers and/or sellers do not have all the necessary information to make the correct decision about a product
  - Asymmetric information is a form of imperfect information where one party (usually the seller) has more or better information than the other (usually the buyer).
    - Information failure, where individuals or firms have inaccurate or incomplete information, resulting in wrong decisions
      - misallocation of resources
- monopoly and monopoly power can lead to market failure
  - monopolist is the sole provider
    - face no direct competition
      - set prices higher than in competitive market
  - price consumers pay does not reflect the true cost
    - inefficiency
  - lack of competitive pressure
    - less incentive to operate efficiently
- immobility of factors of production can lead to market failure
  - Immobility of factors of production
  - Geographical immobility of labour
  - Occupational immobility of labour
  - prevents markets from adjusting efficiently to changes in supply and demand
    - mismatches in resource allocation
- price instability may be a source of market failure
  - price changes by large amounts in short time period
    - signals can be confusing
  - Firms either overproducing or underproducing
  - prices are temporarily high
    - may believe that demand has increased permanently
      - overproduce
        - surpluses
          - wasted resources
            - falling prices
              - resources such as labour and capital are misallocated
  - prices suddenly fall
    - cut back production too much
      - shortages
        - an inefficiently low output

5.6 An inequitable distribution of income and wealth

- difference between income and wealth
  - Income
    - earned income
    - unearned income
  - Wealth
- distinction between equality and equity
  - Equality
    - when everyone has, or is treated, the same
  - Equity
    - fairness
- the market mechanism is likely to result in an unequal distribution of income and wealth.
  - Income (unequal)
    - higher levels of education, training, or skills
      - more in demand in the labor market
        - higher wages
  - Wealth (unequal)
    - inherit substantial assets
      - giving advantage in education, opportunities, and the ability to generate further wealth
    - more assets
      - more income can generate from those assets
        - rich becoming richer
- the market mechanism is likely to result in an unequal distribution of income and wealth.
  - Wealthy individuals
    - have the money to buy expensive luxury
      - market focuses on producing products like luxury
  - people with lower incomes cannot afford basic
    - don't have enough money to create strong demand
      - underproduction of essential goods

5.7 Government intervention in markets

- 5.8 Government failure
  - How government intervention can lead to government failure
    - government intervention in the economy leads to a worse allocation of resources and a fall in welfare
    - net welfare loss
    - evaluation
      - Ultimately, in deciding whether government intervention to correct a market failure should take place, a judgement must be made about whether the size of government failure outweighs the problems caused by the market failure.
  - Sources of government failure
    - inadequate information
      - doesn't have the full information it needs to make informed decisions about how to best allocate resources
      - intervene to soon or too late
    - inappropriate or conflicting objectives
      - often face a trade-off between achieving long term and short term policy objectives
    - Regulation or administration costs can be expensive > benefits
      - worsening of allocation of resources
    - corruption
      - decisions may be influenced by personal gain rather than public interest
  - How government intervention can lead to unintended consequences
    - rice controls may help to solve one problem but creates others by distorting price signals
      - minimum price sends a signal to producers to supply more
        - excess of perishable products
          - waste of resources
      - maximum price sends a signal to producers to supply less
        - excess demand
- market failure provides an argument for government intervention
  - Resources are not put to their best use
  - Welfare is not maximized
  - More or less resources are allocated to some products
  - High prices make goods and services unaffordable to the poor
- How governments influence resource allocation
  - Public expenditure
  - Taxation
  - Regulation
- governments have a range of objectives that affect how they intervene in a mixed economy
  - Interventionist economists
- use of different policies to correct market failure
  - Indirect taxes
    - definition: tax on spending
      - specific: based on quantity, not the price of the good or service
      - Ad valorem: according to value, a percentage of the value of the good or service
    - Taxing negative externalities in production
      - 前面一段描述negative externalities in production的图表 见5.3
        - To achieve the socially optimum
          - tax of AB is needed=external cost of that particular unit
            - tax increases the private cost of the firms of producing this unit
              - only willing to supply this quantity at a price of pt instead of C
                - now taking account of the full social cost
      - new equilibrium price increases to pt
        - equilibrium moves from E to A
          - indirect tax of AB
            - less is sold due to the resulting contraction of demand(qs social optimum)
              - Illustrates the polluter pays principle
    - Taxing negative externalities in consumption
      - 前面一段描述negative externalities in consumption的图表 见5.3
        - To achieve the socially optimum output of qs
          - tax of AB is needed=external cost
            - tax increases the private cost to the firms of producing this unit
              - willing to supply this quantity at a price of pt instead of C
      - new equilibrium price increases to pt
        - equilibrium moves from E to A
          - indirect tax of AB
            - less is sold due to a contraction of demand
              - new equilibrium quantity is qs(social optimum)
    - Advantages and disadvantages of indirect taxes
      - Advantages
        - generate government revenue
        - increase the price of goods or services that produce negative externalities
          - limit harmful activities
      - disadvantages
        - regressive
          - higher burden on low-income individuals
        - demand for a good is inelastic
          - most of the tax will be passed on to the buyers (higher price)
            - increased financial burden
        - difficult to identify "correct" amount
          - too high
            - raise the price too much
              - reduce the quantity below the social optimum
          - too low
            - may not reduce production enough
              - but at least the quantity provided will be closer to the social optimum than before
    - When to Use Indirect Taxes for Market Failure
      - goods with elastic demand
      - alternative goods or services are available to replace
      - combined with other policies
  - Subsidies
    - 前面一段描述positive externalities in consumption的图表 见5.3
      - To achieve the socially optimum output of qs
        - subsidy of AB is needed=external benefit of that particular unit
          - subsidy reduces the private cost to the firms
            - willing to supply this quantity at a price of psub instead of C
      - new equilibrium price falls to psub
        - equilibrium moves from E to B
          - subsidy of AB
            - new equilibrium quantity is qs(social optimum)
              - more is sold
                - extension of demand
    - Subsidising positive externalities in production
      - 前面一段描述positive externalities in production的图表 见5.3
        - To achieve the socially optimum output of qs
          - subsidy of AB is needed=external benefit
            - subsidy reduces the private cost to the firms of producing this unit
              - willing to supply this quantity at a price of psub instead of C
      - new equilibrium price falls to psub
        - equilibrium moves from E to B
          - subsidy of AB
            - new equilibrium quantity is qs(social optimum)
              - extension of demand
    - Advantages and disadvantages of a subsidy
      - advantages
        - improving both efficiency and equity
          - merit goods more affordable
        - long-term economic growth
          - improve social welfare
      - disadvantages
        - financial cost&opportunity cost
          - be expensive for governments
        - waste or environmental issues
          - without fully valuing the product's social benefits
            - overconsumption of the subsidized good
        - Firms may be less incentive
        - demand for a good is inelastic
          - most of the subsidy will be passed onto buyers in the form of a lower price
            - misallocation of resources
              - large subsidy may be needed
        - difficult to identify "correct" amount
          - too high
            - reduce the price too much
              - inefficient use of resources
              - raise the quantity above social optimum
          - too low
            - may not increase production enough
              - but the quantity provided will be closer to the social optimum
  - Price controls
    - minimum price
      - equilibrium price is p where q is sold
        - excess supply of qdqs
          - extension of supply & contraction of demand
            - minimum price is set at a higher level such as pmin
      - with a minimum price
        - firms cannot sell legally at any price lower than pmin
          - consumption will fall from q to qd
            - Consumers are only willing to buy qd at pmin
      - Advantages and disadvantages of a minimum price
        - advantages
          - consistent supply
            - preventing prices from falling too low
          - producers receive a fair income, especially farmers since prices of agricultural products are instable
          - improve public health
            - reduce overconsumption of demerit goods
        - disadvantages
          - Increases the cost for consumers
            - essential goods less affordable for low-income households
          - government may need to buy the surplus
            - maintain the minimum price
            - ↑ public spending
          - government does not buy the excess supply
            - firms may sell the surplus illegally below the minimal price
    - maximum price
      - equilibrium price is p where q is sold
        - excess demand of qdqd
          - extension of demand & contraction of supply
            - maximum price is set at a lower level such as pmax
      - with a maximum price
        - firms cannot sell at any price higher than pmax
          - so consumption will fall from q to qs
      - Advantages and disadvantages of a maximum price
        - advantages
          - Improves Affordability
          - Prevents Monopolistic Exploitation
          - Promotes Social Equity
            - more people can access essential goods regardless of income level
        - disadvantages
          - Shortages
            - supply decrease
              - producers may not find it profitable to supply
          - Black Markets
  - Buffer stocks
    - S1 → S2
      - favourable weather conditions
        - p1 no action is needed
    - left to the market
      - price fall below the price floor to p2
        - intervene
          - buying wheat to increase demand
            - raise the price
              - consumers are willing to buy q2
                - At p3 according to the market demand curve D1
                - price will rise above price floor to p3
                - extra quantity q2q3 must be bought to achieve an equilibrium price of p3
    - reduce supply of wheat from S1 to S2
      - unfavourable weather
    - if left to the market
      - price rise above the price ceiling to p2
        - need to intervene
          - selling wheat to increase supply to S3
            - according to the market supply curve S2, firms are willing to sell q2
              - extra quantity q2q3 must be sold to achieve equilibrium price of p3
              - price fall below the price ceiling to p3
    - Advantages and disadvantages of a buffer stock scheme
      - advantages
        - stables price for consumers and income for producers
      - disadvantages
        - difficult to know where to set price ceiling and floor
        - costs to buy extra supplies
        - products cannot be stored for too long
        - run out of stocks if there are a few harvests
  - Pollution permits
    - Advantages and disadvantages of pollution permits
      - advantages
        - Reduce pollution over time
        - have the flexibility to choose how to reduce emissions
          - sell excess permits
            - reduce emissions cheaply
              - facing higher costs will buy permits
                - more efficient allocation of resources & desired reduction in pollution at the least cost
      - disadvantages
        - requires complex monitor
        - uncertainty
          - price of permits can fluctuate
        - May not apply to all geographical areas
  - Extension of property rights
    - advantages
      - Force the producer of the external cost to pay the person or firm affected
      - more efficient use
        - encourage individuals and businesses to invest in and develop resources
      - reduce tragedy of commons
        - individuals are more likely to manage resources sustainably
    - disadvantages
      - difficult to extend property rights to some resources(air)
      - exacerbating inequality
        - benefit wealthy or powerful groups
      - benefit fianancially from other paying for their use
        - patent
  - State provision
    - necessary if there is a missing market
    - large financial and opportunity cost
    - difficult to know what quantity to supply to achieve the social optimum
    - does not guarantee that it will be consumed
  - Regulation
    - Regulation needs to be backed by penalties for those who break the laws.
    - some people may believe the government is being too paternalistic
    - It is easy to understand
    - otherwise, may lead to black market
      - need to be monitored and enforced
  - Other solutions
    - reduce information failure
      - advertise or promote certain activities
    - narrowing gap between rich and poor
      - progressive income taxes
    - provision of merit goods

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